

Global Renewable Energy in 2025

How Far Have We Come?

Fossil fuels still drive ~75% of GHG emissions

At least 5 million premature deaths/year from air pollution

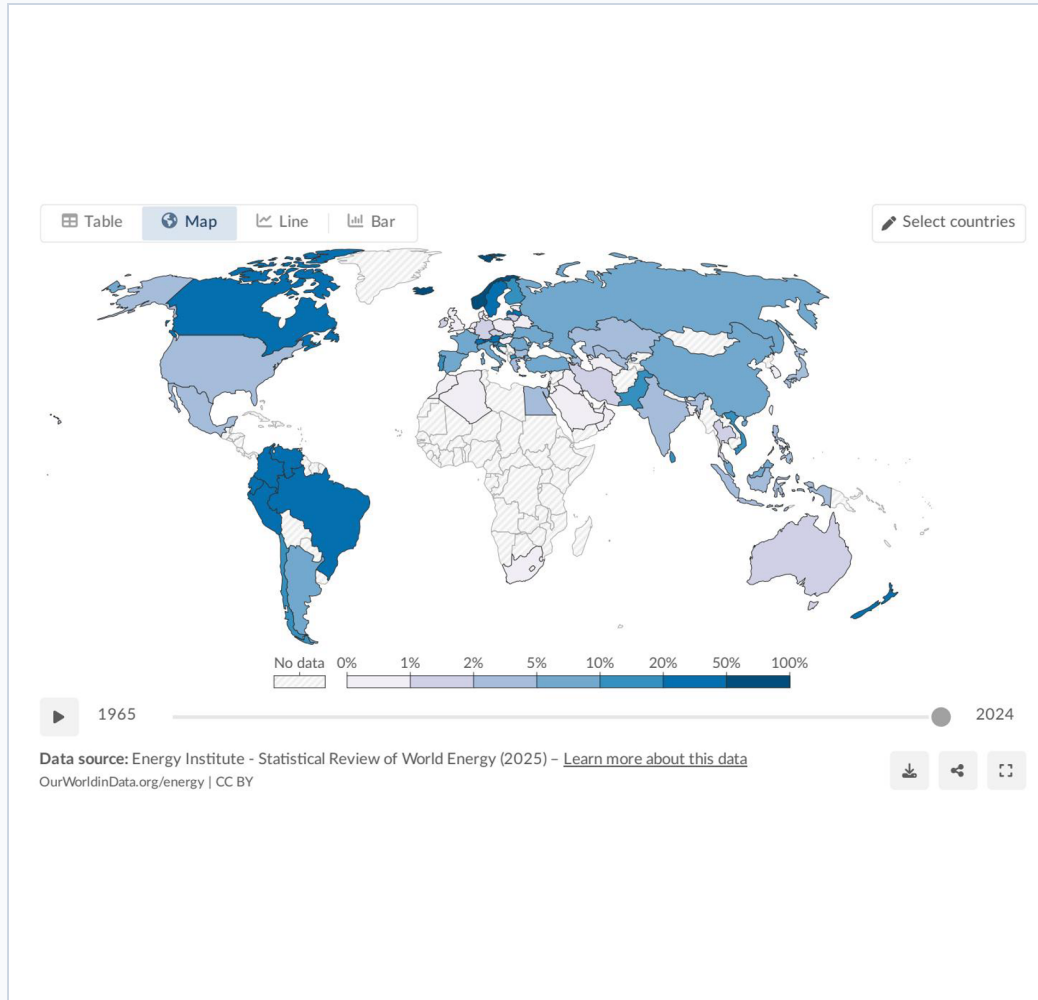
Audience: Policy analysts and energy sector strategists

Focus: Progress, technology mix, and regional disparities

Source: Our World in Data (Ember 2026; Energy Institute 2025)

~14% of Global Primary Energy Is Renewable (2024)

Primary energy includes electricity, transport, and heating — where decarbonization remains uneven.



Key insight

Renewables are stronger in electricity than in total energy demand.

- Leaders (e.g., Norway, Brazil, New Zealand) show high primary-energy renewable shares.
- Many fossil-fuel producers remain below 5%.

Implication: Electrification pace is critical.

~30% of Global Electricity Is Renewable (2025)

Electricity is the leading edge of the transition — but country performance differs sharply.

Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.

Our World
in Data

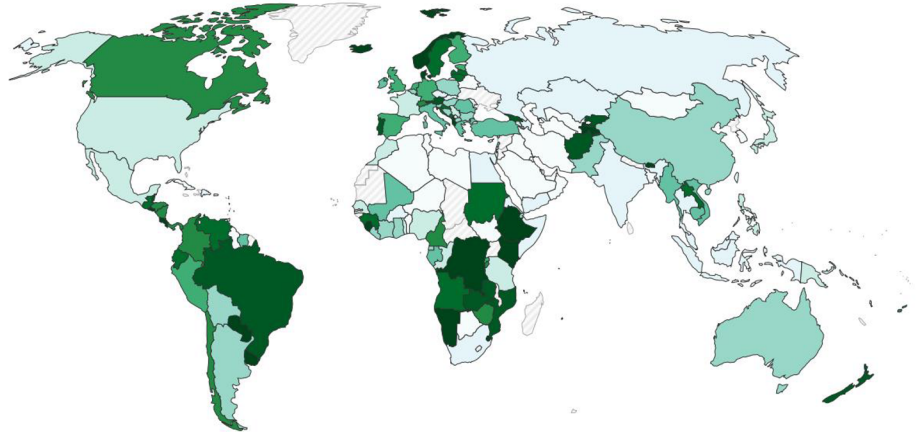
Table

Map

Line

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Select countries



No data 0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%

1985 2025

Data source: Ember (2026); Energy Institute - Statistical Review of World Energy (2025) - [Learn more about this data](#)

OurWorldinData.org/energy | CC BY

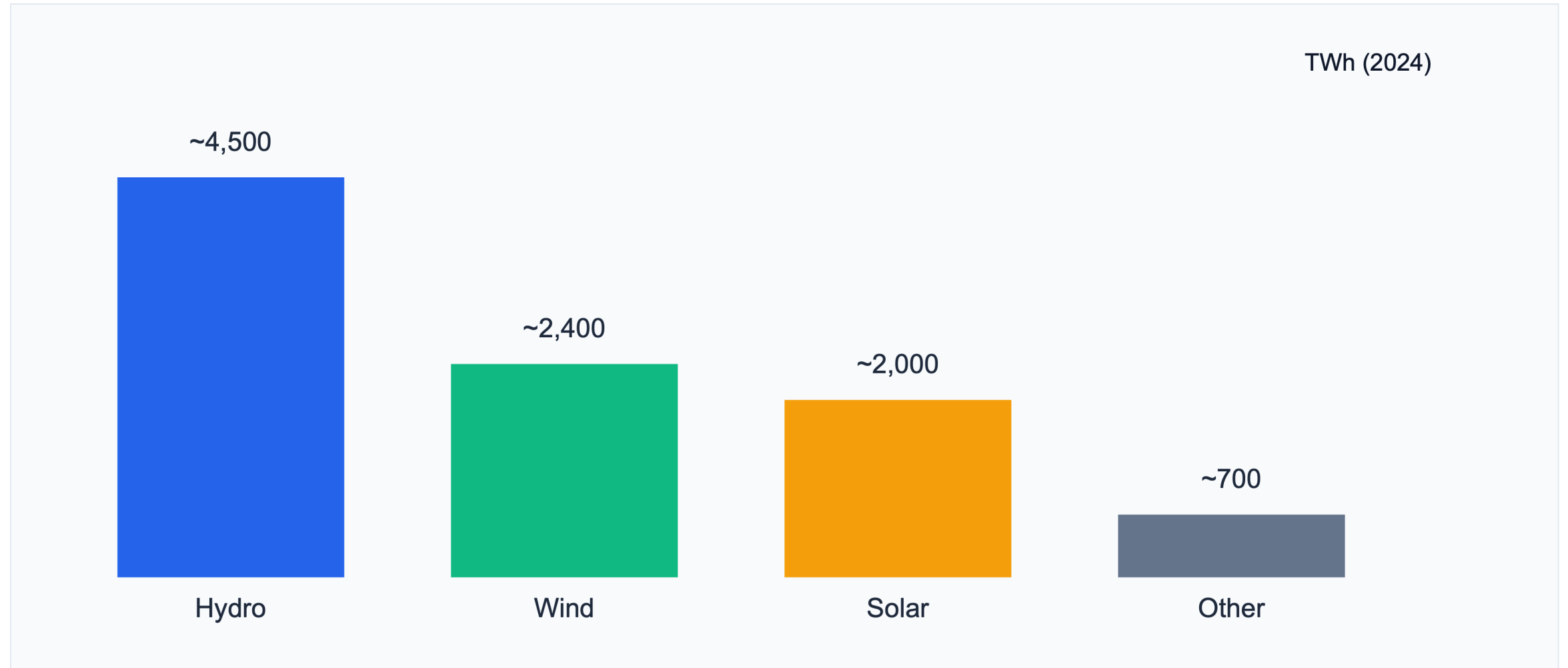
Selected Renewable Electricity Shares

Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%

World average: ~30%

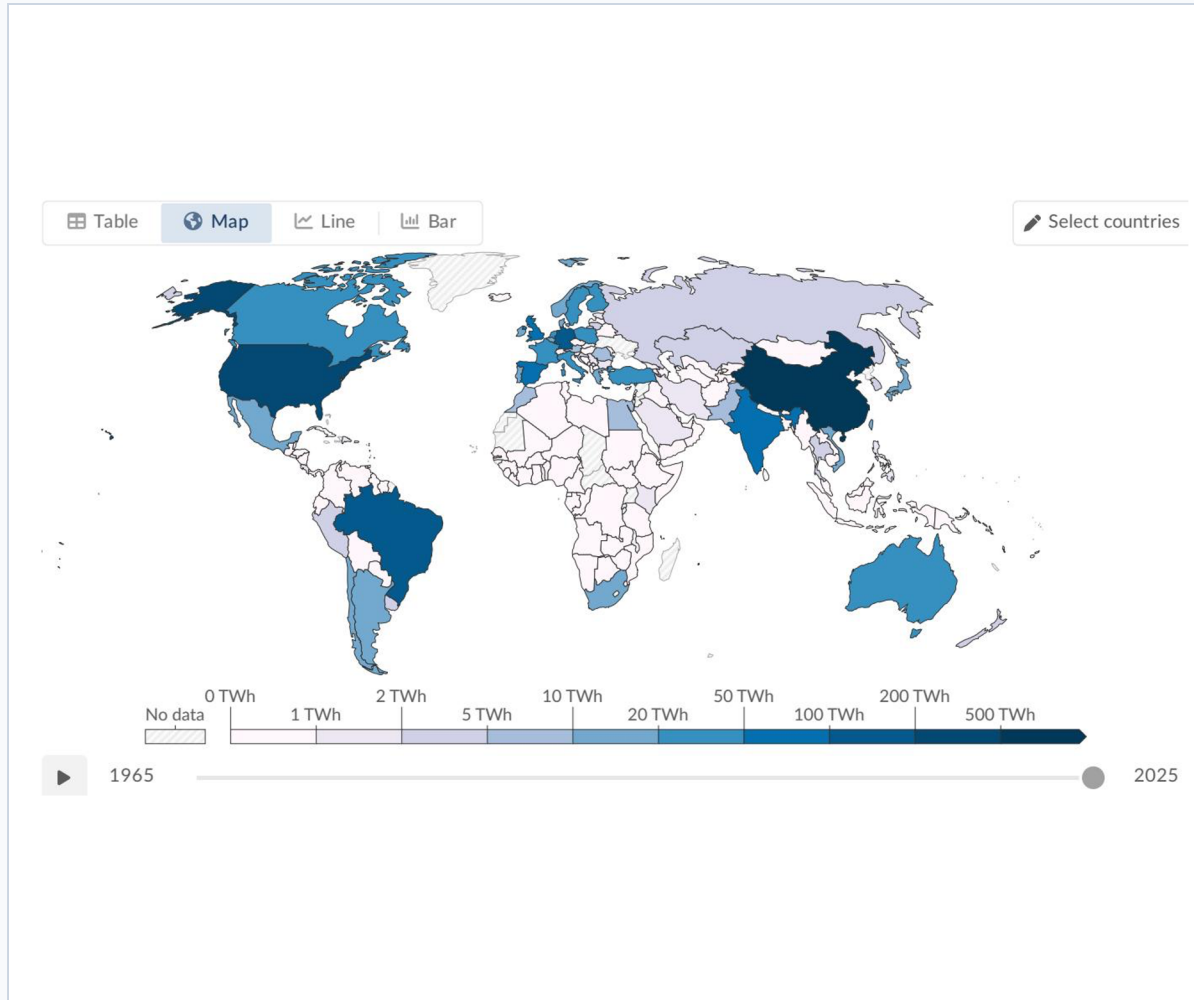
Hydro ~4,500 TWh Still Leads Renewable Generation

Wind and solar are rapidly closing the gap in global electricity output.



Hydropower Provides ~47% of Renewable Electricity

Hydro remains foundational, with major output concentrated in a few large producers.

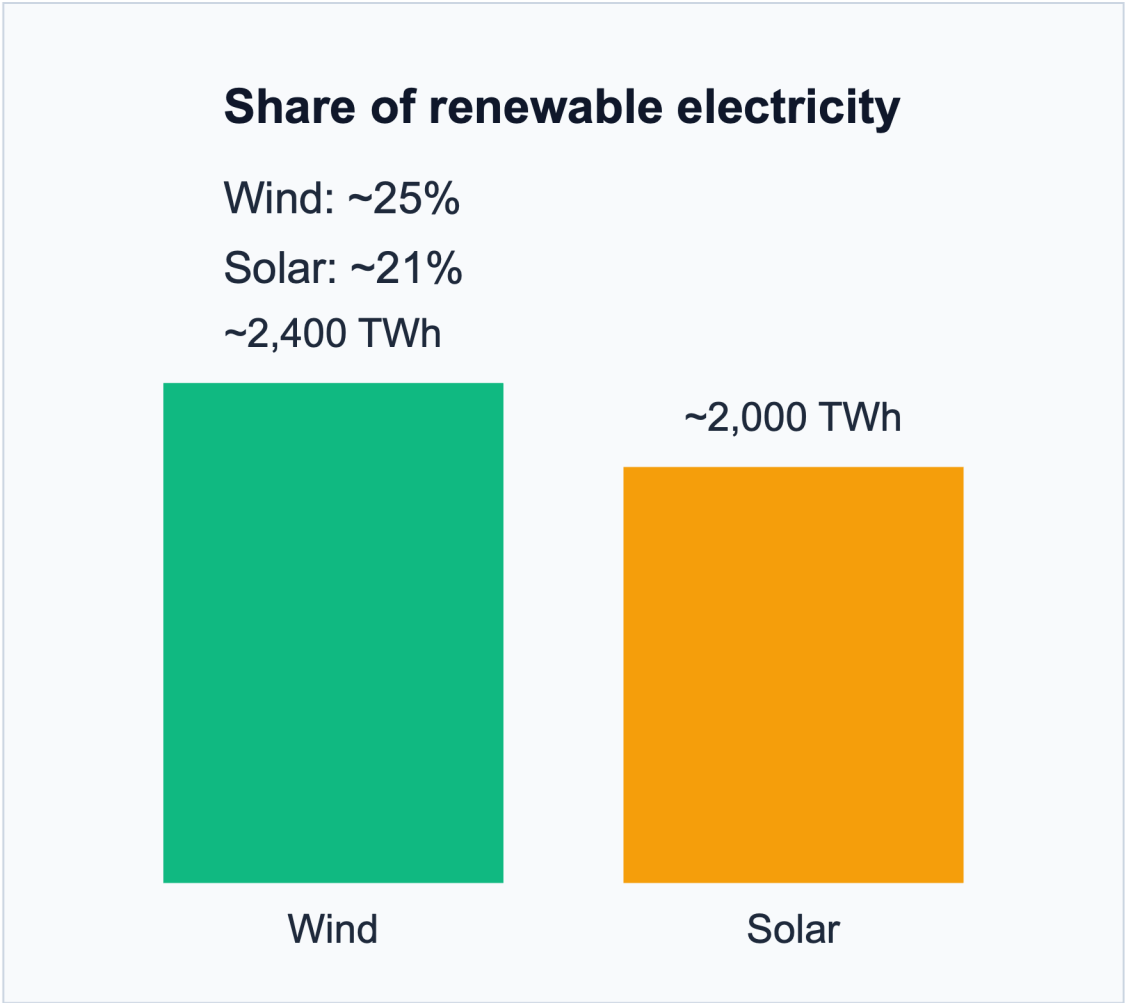


Strategic implications

- Hydro is mature and dispatchable.
- Expansion constrained by geography, permitting, and ecological concerns.
- Top producers include China, Brazil, and Canada.

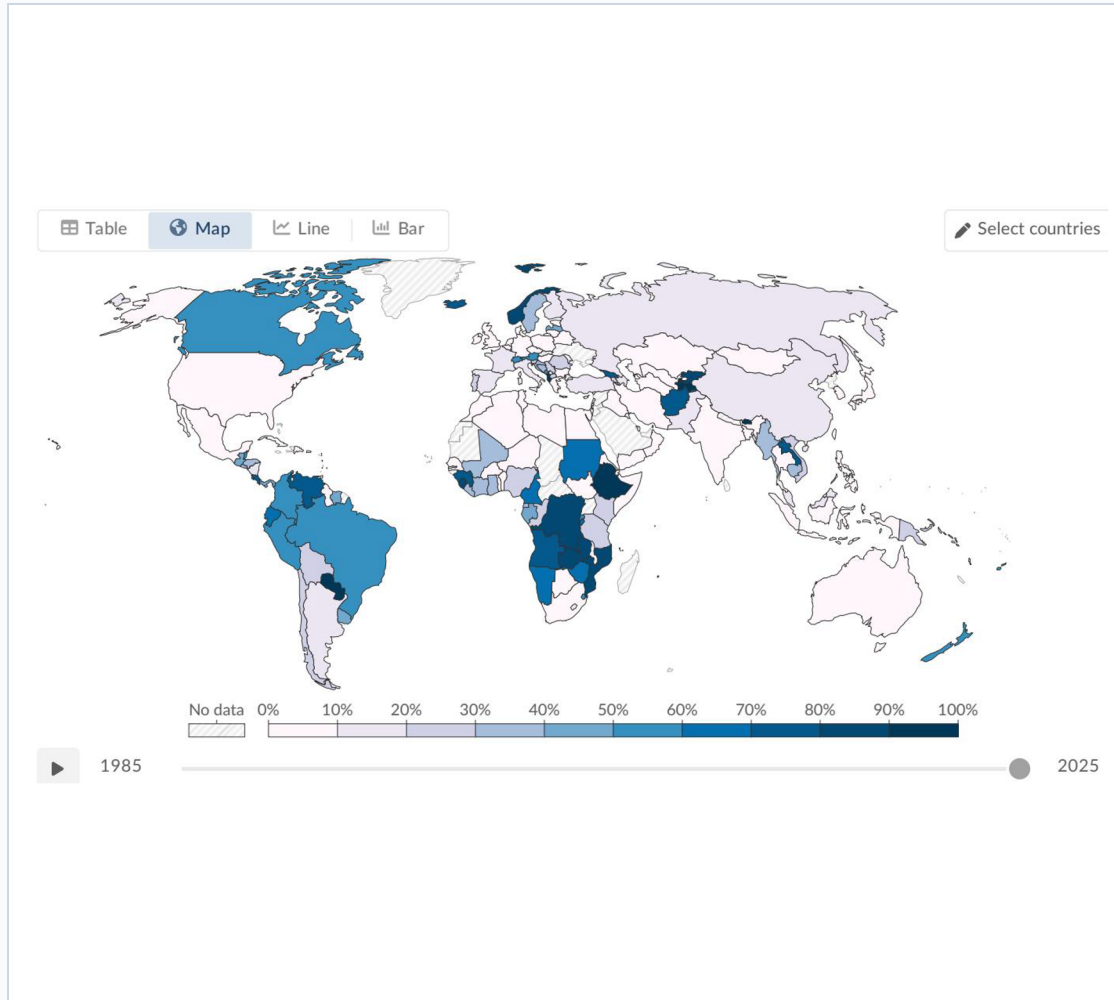
Wind and Solar Are the Fastest-Growing Sources

Generation-based comparison (2024): wind ~2,400 TWh (25%) vs solar ~2,000 TWh (21%).



Regional Gaps Persist: Many Countries Remain Below 10%

Leaders rely on hydro or sustained wind/solar policy; laggards remain fossil dependent.



Observed pattern

- Scandinavia + Brazil + Canada: high renewable shares, hydro strength.
- Denmark/Germany: wind and solar acceleration from policy investment.
- Parts of Sub-Saharan Africa and Middle East remain below 10%.

Key Findings: Progress Is Real, But Insufficient

Global transition momentum is clear — system-wide decarbonization still lags what climate goals require.

- 1) Renewables now supply ~14% of primary energy and ~30% of electricity.**
- 2) Hydropower remains largest (~47%), but growth is slower.**
- 3) Wind + solar nearly match hydro and are growing fastest.**
- 4) Regional disparities remain severe (<10% to ~98%).**
- 5) Transport and heating are the key bottlenecks.**

Call to Action: Accelerate the Next Phase

Electricity progress proves scaling works — now extend that pace across entire energy systems.

Strategic priorities for 2026–2030

- Scale grid investment and flexibility for variable renewables.
- Speed policy support in low-share regions (<10%).
- Drive electrification of transport and heating demand.
- Maintain affordability and reliability during transition.

Bottom line: momentum exists — the pace must now multiply.